

Module/Pathway/Taxon Review: De novo biotin biosynthesis via the BioC–BioH route in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556)
Bucket: KEGG `ppu00780` "Biotin metabolism"; module area = de novo biotin biosynthesis (BioC–BioH / modified-FAS route) **Review date:** 2026-07-17 · Evidence: literature + UniProtKB + InterPro/Pfam + KEGG (no experimental data files supplied)

1. Executive summary

The biotin BioC–BioH module is **fully satisfiable** in *P. putida* KT2440. All six pathway-specific enzymatic steps are encoded by candidate genes, five of them in a single contiguous `bioBFHCD` operon (PP_0362–PP_0366) with `bioA` (PP_4984) unlinked. Each of the six core genes carries a **diagnostic, biotin-specific InterPro/Pfam signature** (not merely a propagated gene name), and KEGG assigns all six route-defining KOs (K01012, K00652, K02170, K02169, K01935, K00833). The organism is a predicted **biotin prototroph** using the canonical modified-fatty-acid-synthesis route; the lineage-specific alternatives (BioI/BioW/BioU) are absent and **not expected**.

The chief curation issues are **over-inclusion**, not gaps: 9 of the 16 bucket genes are **generic fatty-acid-synthase (FAS) enzymes** that KEGG pulls into `ppu00780` because the pimelate chain is built by shared FAS chemistry — these are explicit dependencies, not biotin-specific members — and one gene (`PP_2783`, a p-cumate catabolic dehydrogenase) is a **spurious inclusion**. The single genuine biotin-specific gene warranting closer scrutiny is `bioH` (**PP_0364**): it is the least-conserved step (UniProt PE4/Predicted, no assigned EC in UniProt), though its membership in the dedicated BioH carboxylesterase InterPro family (IPR050228) supports the annotation.

Evidence for KT2440 is **homology/domain/pathway-database based (PE3–4)**; there is no direct biochemical or genetic characterization of the KT2440 bio proteins themselves. The strongest transferable experimental evidence comes from *P. aeruginosa* (operon-encoded BioH; strong transfer within *Pseudomonas*) and from *E. coli* (mechanism of the whole route; strong for conserved steps).

2. Target-organism pathway definition

Process included (module boundary): de novo synthesis of the biotin cofactor from a malonyl-thioester precursor, comprising two conceptual halves:

1. **Pimelate (C7 α,ω -dicarboxylate) moiety synthesis by a "disguised" FAS pathway.** BioC O-methylates the free ω -carboxyl of malonyl-ACP so that the atypical substrate is accepted by the housekeeping FAS elongation enzymes; two elongation cycles yield pimeloyl-ACP methyl ester, which BioH hydrolyzes to pimeloyl-ACP + methanol (Lin, Hanson & Cronan 2010,

P 20693992

Cronan & Lin 2011,

P 21435937

Lin & Cronan 2012,

P 22965231

2. **Assembly of the fused rings:** BioF (pimeloyl-ACP + L-Ala $\rightarrow$ 8-amino-7-oxononanoate, KAPA), BioA (KAPA + SAM $\rightarrow$ 7,8-diaminononanoate, DAPA), BioD (DAPA + CO₂/ATP $\rightarrow$ dethiobiotin), BioB (dethiobiotin + S $\rightarrow$ biotin).

Kept separate (neighboring processes / overview maps): - **Fatty-acid biosynthesis (KEGG map00061 / fab genes).** The FAS elongation enzymes (FabB/F/G/Z, enoyl-ACP reductase) act on the BioC-masked intermediate but are **not biotin-specific**; they belong to FAS and should be modeled as a dependency, not module members. - **Biotin activation / protein biotinylation and regulation (BirA).** Per the module scope, biotin-[acetyl-CoA-carboxylase] ligase activity and BirA-mediated transcriptional repression are **outside the biosynthesis boundary**. - **Aromatic/p-cumate catabolism** (source of the mis-bucketed **PP_2783**).

Alternate names / database definitions: MetaCyc "biotin biosynthesis I (BioC-BioH)"; KEGG module **M00123** "Biotin biosynthesis, pimeloyl-ACP/CoA => biotin"; the pimelate-initiation genes `bioC`/`bioH` correspond to KOs **K02169**/**K02170**. Distinct KEGG routes not used here: **M00573** BioI (P450, K16593), **M00577** BioW (pimeloyl-CoA synthetase, K01906), **M00950** BioU (K25570).

3. Expected step model and satisfiability

#	Step (substrate → product)	Enzyme / KO / EC	Candidate gene	Status
1	malonyl-ACP → malonyl-ACP methyl ester	BioC O-MTase / K02169 / 2.1.1.197	bioC PP_0365	covered
2	pimeloyl-ACP methyl ester → pimeloyl-ACP	BioH esterase / K02170 / 3.1.1.85	bioH PP_0364	covered (verify; least-conserved)
3	pimeloyl-ACP → KAPA (8-amino-7-oxononanoate)	BioF AONS / K00652 / 2.3.1.47	bioF PP_0363	covered
4	KAPA → DAPA (7,8-diaminononanoate)	BioA aminotransferase / K00833 / 2.6.1.62	bioA PP_4984	covered
5	DAPA → dethiobiotin	BioD synthetase / K01935 / 6.3.3.3	bioD PP_0366	covered
6	dethiobiotin → biotin	BioB radical-SAM synthase / K01012 / 2.8.1.6	bioB PP_0362	covered

Dependency (not a module step): FAS elongation of the masked intermediate (steps between 1 and 2) — supplied by housekeeping FabB/FabF/FabG/FabZ/enoyl-ACP-reductase + ACP + malonyl-ACP. Present in KT2440 but should be modeled as an external dependency.

Not expected in this taxon: BioI (P450) pimelate route, BioW (pimeloyl-CoA) route, BioU route — mark `not_expected_in_target_taxon`.

All six expected steps are encoded → **module = COVERED / satisfiable**.

4. Candidate genes and evidence

Core biotin-specific genes (promote/retain)

- bioC — PP_0365 / Q88QW9.** Malonyl-ACP O-methyltransferase (EC 2.1.1.197, K02169). InterPro IPR011814 **BioC** + IPR050602 Malonyl-ACP_OMT; Pfam Methyltransf_11 (SAM-dependent). UniProt PE3, "Biotin biosynthesis" keyword. Operon position (...HCD...). *Evidence:* homology + diagnostic domain. Confidence **high**. Caveat: mechanism proven in *E. coli/B. cereus*
 (P 22965231),
 transfer to KT2440 strong (conserved family).
- bioH — PP_0364 / Q88QX0.** Pimeloyl-ACP-methyl-ester esterase (EC 3.1.1.85, K02170). InterPro IPR050228 **Carboxylesterase_BioH**, α/β -hydrolase (Pfam Abhydrolase_6). **UniProt PE4 (Predicted), no EC assigned in UniProt, no biotin keyword.** *Evidence:* homology/synteny **plus intact catalytic machinery** — the 243-aa protein has the AB_hydrolase-1 domain (res 5–231), a G-X-S-X-G nucleophile elbow (**GWSLG, Ser68**) and a C-terminal catalytic Asp/His (...GMVE-**D**-SS-**H**-AF...), reproducing the Ser–Asp–His triad of characterized BioH esterases (*E. coli* / *K. pneumoniae* Ser82–Asp207–His235;
P 31615653).
 Confidence **moderate→high**. Caveat: the esterase step is the "wild card" of the pathway — in different bacteria BioH is replaced by non-homologous α/β -hydrolases (BioG/BioJ/BioK/BioV;
P 28526858).
 Here it is operon-encoded like *P. aeruginosa* BioH, and the catalytic triad is complete, which supports the call; residual uncertainty is only substrate specificity, best resolved by a Δ bioH complementation test.
- bioF — PP_0363 / Q88QX1.** 8-amino-7-oxononanoate synthase (EC 2.3.1.47, K00652). InterPro IPR022834 **AONS_Proteobacteria** (matches expected proteobacterial BioF clade); PLP-dependent Aminotran_1_2. PE3, biotin keyword. Confidence **high**.
- bioA — PP_4984 / Q88D44.** Adenosylmethionine–KAPA aminotransferase (EC 2.6.1.62, K00833). InterPro IPR005815 **BioA**, Aminotran_3 (SAM as amino donor — the standard proteobacterial route). PE3, biotin keyword. Unlinked from the operon (common). Confidence **high**.

- **bioD** — PP_0366 / Q88QW8. ATP-dependent dethiobiotin synthetase (EC 6.3.3.3, K01935). InterPro IPR004472 **DTB_synth_BioD**, P-loop NTPase. PE3, biotin keyword. Confidence **high**.
- **bioB** — PP_0362 / Q88QX2. Biotin synthase (EC 2.8.1.6, K01012). InterPro IPR024177 **Biotin synthase**, Radical_SAM + BATS domains. PE3, biotin keyword. Confidence **high**. (Requires Fe-S cluster machinery + SAM; sulfur donor debated but housekeeping.)

Regulatory / boundary gene

- **birA** — PP_0437 / Q88QQ1. Bifunctional biotin-protein ligase / biotin operon repressor (EC 6.3.4.15, K03524). InterPro IPR030855 Bifunct_BirA (BPL_LplA_LipB + HTH_11). Genuine, but **outside the biosynthesis module boundary** (activation + regulation). Retain in bucket for context; do not count toward biosynthesis satisfiability.

Generic FAS / dependency genes (over-propagated into the bucket)

fabB PP_4175 (K.-ACP synthase I, 2.3.1.41), fabF PP_1916 & PP_3303 (KAS-II, 2.3.1.179), fabG PP_1914 (KR, 1.1.1.100), fabZ PP_1602 (dehydratase, 4.2.1.59), PP_1852 (enoyl-ACP reductase, 1.3.1.10), PP_0581 (3-oxoacyl-ACP reductase), PP_2540 (SDR-family oxidoreductase). **None carries a biotin keyword**. These are housekeeping FAS enzymes that operate on the BioC-masked intermediate. Curation: **dependency / not biotin-specific**; do not treat as module members.

Spurious inclusion

- PP_2783 / Q88J66. 2,3-dihydroxy-2,3-dihydro-p-cumate dehydrogenase (EC 1.3.1.58) — **p-cumate/aromatic catabolism**, no biotin role. Clear over-annotation artifact of the KEGG bucket; **remove** from the biotin module.

5. Gaps, ambiguities, and likely over-annotations

- **No true gaps**. Every expected step has a candidate gene with a diagnostic domain; pimelate initiation is not a gap because bioC+bioH are present.
- **bioH ambiguity (only real uncertainty)**. Least-conserved step; PE4/Predicted; no UniProt EC. Family membership (IPR050228) and operon context support it, but functional confirmation is lacking for KT2440. → candidate_uncertain-leaning-covered; promote to full review.

- **Broad EC on bioH metadata.** Local metadata lists both EC 3.1.1.1 (generic carboxylesterase) and EC 3.1.1.85 (specific pimeloyl-ACP-methyl-ester esterase). Prefer the **specific** 3.1.1.85 for the module; 3.1.1.1 is over-general and could over-propagate.
- **Paralog notes.** `fabF`/KAS-II activity is represented by two loci (PP_1916, PP_3303) — normal FAS redundancy, irrelevant to biotin-specific curation. Any of several FAS reductases could service the masked intermediate; do not single one out.
- **FAS over-inclusion (9 genes).** Bucket artifact from KEGG map topology; flag as dependency.
- **PP_2783 mis-bucket (1 gene).** Remove.

6. Module and GO-curation recommendations

Per-step status: - Steps 1–6 (bioC, bioH, bioF, bioA, bioD, bioB): **covered** — with bioH annotated "covered, verify" (moderate confidence). - FAS elongation between steps 1–2: model as **external dependency** (fatty-acid biosynthesis), not a biotin module step. - BioI / BioW / BioU alternative routes: **not_expected_in_target_taxon**.

Module-level: - Mark the module **satisfiable/COVERED** for *P. putida* KT2440. - **module_needs_revision (bucket cleanup):** remove the 9 generic FAS genes and `PP_2783` from the biotin-specific member set; retain them only as an explicit FAS dependency link. Move `birA` to a "regulation/activation" adjunct outside the biosynthesis boundary. - Generic module boundaries are **appropriate** for this organism (six-enzyme model fits); the KEGG *bucket* is what is inflated, not the module definition.

GO curation: no new GO terms required — existing terms suffice: GO:0009102 biotin biosynthetic process; GO:0010340 carboxyl-O-methyltransferase (BioC); GO:0090499 pimeloyl-ACP methyl ester carboxylesterase / GO:0052689 (BioH); GO:0008710 8-amino-7-oxononanoate synthase (BioF); GO:0004015 adenosylmethionine-8-amino-7-oxononanoate transaminase (BioA); GO:0004141 dethiobiotin synthase (BioD); GO:0004076 biotin synthase (BioB). Recommend annotating KT2440 bio genes with evidence code **IEA/ISS** (inferred from sequence/homology), not IDA, since no direct assays exist for this strain.

7. Genes to promote to full fetch-gene review

1. **bioH (PP_0364)** — highest priority. Non-conserved esterase step; PE4/Predicted; UniProt lacks EC. Confirm it is the functional pimeloyl-ACP-methyl-ester esterase (vs. a substitute α/β -hydrolase) and resolve the broad-vs-specific EC. Ideal experiment: complementation of an *E. coli* Δ bioH mutant.
2. **bioC (PP_0365)** — secondary. Confirm malonyl-ACP (not malonyl-CoA) specificity and that no second SAM-MTase confounds annotation.
3. **bioB (PP_0362)** — light-touch. Verify Fe-S/radical-SAM residue conservation and available sulfur-donor context (housekeeping, but worth a domain-completeness check).

bioF, bioA, bioD are high-confidence and need only routine confirmation.

8. Key references

- Lin S, Hanson RE, Cronan JE. *Biotin synthesis begins by hijacking the fatty acid synthetic pathway*. Nat Chem Biol 2010.

P 20693992

(Defines the BioC-methylation / BioH-demethylation route.)

- Cronan JE, Lin S. *Synthesis of the α,ω -dicarboxylic acid precursor of biotin by the canonical fatty acid biosynthetic pathway*. Curr Opin Chem Biol 2011.

P 21435937

(Pimelate via modified FAS; BioI as the restricted alternative.)

- Lin S, Cronan JE. *The BioC O-methyltransferase catalyzes methyl esterification of malonyl-ACP, an essential step in biotin synthesis*. J Biol Chem 2012.

P 22965231

(BioC substrate = malonyl-ACP.)

- Cao X, Zhu L, Hu Z, Cronan JE. *Expression and Activity of the BioH Esterase of Biotin Synthesis is Independent of Genome Context*. Sci Rep 2017.

P 28526858

(*P. aeruginosa* operon-encoded BioH; esterase "wild-card" plasticity — strongest within-*Pseudomonas* transfer.)

- Wang M et al. *Structural insight into the carboxylesterase BioH from Klebsiella pneumoniae*. 2019.

P 31615653

(BioH catalytic triad / α/β -hydrolase fold.)

- Chow J et al. *The Thaumarchaeon N. gargensis carries functional bioABD genes and a promiscuous Δ bioH-complementing esterase EstN1*. 2018.

P 30218044

(Illustrates esterase substitutability.)

- Zeng J et al. *Pimeloyl-ACP methyl ester esterase...* 2020.

P 32117167

- Data resources: UniProtKB proteome UP000000556; InterPro/Pfam; KEGG `ppu00780`, modules M00123/M00573/M00577/M00950.

Evidence grade summary

Claim	Basis	Directness for KT2440
Six steps encoded; operon <code>bioBFHCD</code>	UniProt loci + KEGG	Direct genomic (strain)
Diagnostic family assignments	InterPro/Pfam	Direct sequence (strain)
Route = BioC-BioH (not BioI/W/U)	KEGG KO presence/absence	Direct genomic (strain)
Mechanism of each step	<i>E. coli</i> / <i>B. cereus</i> biochemistry	Inferred (strong for conserved steps)
bioH is a functional esterase	<i>P. aeruginosa</i> / <i>E. coli</i> + domain family	Inferred (moderate; verify)
Prototrophy	pathway completeness	Inferred (no growth assay cited)

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